✅ Project Title: Pharma Drug Information Advisor

□ Problem Statement

In the pharmaceutical industry, researchers, medical advisors, and regulatory professionals often struggle to quickly locate and understand critical information from vast clinical trial reports, FDA guidelines, and drug research documents. Traditional keyword search is inefficient and often misses contextual relevance.

To solve this, we propose **"Pharma Drug Information Advisor"** — an **Agentic AI + RAG-powered assistant** that can:

- Dosage Guideline
- Drug Interaction
- Regulatory Compliance

This is achieved by combining **retrieval-augmented generation (RAG)** with **AI agents**, allowing the system to *fetch*, *reason*, and *respond* based on real-world documents.

□ Document Store

- Accepted formats: PDF, HTML, or TXT
- Sample sources:
    - Clinical trial documents (CT.gov exports)
    - FDA approval letters
    - Drug research PDFs

□ RAG Pipeline Architecture

1. **Document Ingestion & Chunking**

  - Load and parse all uploaded documents using LangChain-compatible loaders.
  - Split content into overlapping chunks for better context preservation using `RecursiveCharacterTextSplitter`.

2. **Embedding & Vector Storage**
  - Use **Google Generative AI Embeddings** (`models/embedding-001`) to generate embeddings.
  - Store vectors using **Pinecone or chroma db** for fast and scalable similarity search.

3. **Query Retrieval**
  - User submits a drug-related question via UI.
  - System retrieves top-k relevant chunks from the FAISS vector DB.

4. **Response Generation**
  - Retrieved context is passed to **Gemini 1.5 Flash** for generating a precise, domain-aware answer.
  - Gemini generates answers optimized for medical clarity and regulatory correctness.

□ Agentic AI Workflow (LangGraph)

- **Agent A: Drug Research Expert**
  - Accepts queries and triggers appropriate sub-agents/tools.
- **Tool: Document Fetcher Tool**
  - Retrieves top-matching vector chunks.
- **Gemini 1.5 Flash**
  - Used as the reasoning engine to compose accurate responses.

- **LangGraph Framework**
    - Defines the multi-agent, parallel, and retry-based flow.
    - Enables conditional logic and role-based execution.

□ Impact

- Reduces 80% manual effort in locating critical regulatory data.
- Increases accuracy in interpreting multi-format drug documents.
- Enables real-time insights during drug development and approval processes.